

39) A detective has interviewed four witnesses to a crime. From the stories of the witnesses the detective has concluded that if the butler is telling the truth then so is the cook; the cook and the gardener cannot both be telling the truth; the gardener and the handyman are not both lying; and if the handyman is telling the truth then the cook is lying. For each of the four witnesses, can the detective determine whether that person is telling the truth or lying? Explain your reasoning.

Soln: p : Butler is telling the truth q : Cook is telling the truth
 r : Gardener is telling the truth s : Handyman is telling the truth

① $p \Rightarrow q$ is True ② $\neg(q \wedge r)$ is True ③ $\neg(\neg r \wedge \neg s)$ is True
 ④ $r \Rightarrow \neg q$ is True. By equivalence of contrapositive, $\neg(\neg q) \Rightarrow \neg r$ is True

~~not~~ We know that $q \Rightarrow \neg(\neg q)$ is True. ⑤

Let, p be True. $\therefore$ From ①, and defn. of conditional, q is True. From ⑤, and defn. of bi-conditional, $\neg(\neg q)$ is True. From ④ and defn. of conditional, $\neg r$ is True. By defn. of negation, r is False.

By defn. of AND, $q \wedge r$ is False. By defn. of negation, $\neg(q \wedge r)$ is True. (Consistent with ②).

Since $\neg(\neg r \wedge \neg s)$ is True, by defn. of negation, $\neg r \wedge \neg s$ is False. Since $\neg r$ is True, by defn. of AND, $\neg s$ is False. By defn. of negation, s is True.

$\therefore$ If p is True, then q and s are True, r is False.

Soln 1: Butler, Cook, Hand (Wrong Solution)

Soln: p : Butler is telling the truth q : Cook is telling the truth
 r : Gardener is telling the truth s : Handyman is telling the truth

① $p \Rightarrow q$ is True ② $\neg(q \wedge r)$ is True ③ $\neg(\neg r \wedge \neg s)$ is True
 ④ $s \Rightarrow \neg q$ is True. By equivalence of contrapositive, $\neg(\neg q) \Rightarrow \neg s$ is True
 We know that $q \Rightarrow \neg(\neg q)$ is True. ⑤